

Codfishes and relatives

PHYLUM	Chordata
CLASS	Osteichthyes
SUBCLASS	Actinopterygii
SUPERORDER	Paracanthopterygii
ORDERS	5
FAMILIES	24
SPECIES	About 667

Codfish, or cods, are a large and important group of mainly predatory fish, which are also vital prey for many larger species, such as sharks. As well as cod itself, true codfish—from the order Gadiformes—include haddock, hake, pollack, and many other commercially important species, as well as the widespread deep-water grenadiers. Other

distant relatives in the same superorder include freshwater trout-perches and dories. Apart from them, the majority are marine fish with a small number found in fresh or brackish water. The largest variety is found in the Atlantic, where many species live and feed just above the sea bed, often in very large shoals. Grenadiers are found at depths of least 20,000 ft (6,000 m) and have been filmed around hydrothermal vents.

Anatomy

True codfish typically have a long, narrow body, with spineless fins that are supported by soft rays. The paired pelvic fins are positioned well forward on the underside, behind the head and right below or forward of the pectoral fins. They have 1, 2, or 3 dorsal fins on their back but members

of the cod family (Gadidae) always have 3. Many have a single sensory barbel on the lower jaw, which helps bottom-feeding species detect hidden invertebrate prey. Species such as hake that feed in mid-water do not have a barbel, but have a protruding lower jaw and large teeth suitable for catching slippery fish and squid.

Grenadiers have a disproportionately large head and a body that tapers to a thin, pointed, ratlike tail, which gives them their alternate names of rattails or ratfish. They spend most of their time near the seabed, hunting and scavenging, and are very unfussy feeders. There are several hundred different species around the world and they are among the most successful of all deep-water fish. The confusingly named trout-perches are found in fresh water in the US. They are so called because the two true species (in the family Percopsidae) have a dorsal fin with spines at the front, like that of perches, and a small adipose fin on the back near the tail, as trout do. Grouped with them in the same order (Percopsiformes) are the specialized cavefishes, which only have vestigial eyes and tiny pelvic fins, or none at all.

Dories (in the order Zeiformes) do not resemble codfish but there is molecular evidence that shows they are related to codfish. They have deep but thin bodies, which is obvious from the side but not easy to see head-on. The front part of the dorsal fin has strong spines, as does the anal (belly)

fin in some species. Most have a large mouth that can be extended to suck in prey. They live in the deep sea.

Life cycle

A fully mature Atlantic cod can spawn as many as 10 million eggs in a single season and many other large species of codfish are similarly prolific egg layers. While a few bottom-living codfish species, such as rocklings, lay their eggs in hidden crevices in the sea bed, most simply broadcast them into the water. Losses of eggs and fry to predators are therefore huge.

Establishing exactly where codfish species spawn is difficult, but it is known that cod and other codfish do migrate to local spawning grounds. These are often in the vicinity of shallow, sheltered nursery sites, such as estuaries, where juveniles can grow in relative safety. Cod and many other codfish are heavily exploited and many stocks have been badly overfished, leading to the collapse of some. Little is known about the life cycle of grenadiers. They produce in the order of 100,000 tiny, buoyant eggs, which are thought to float up and develop in mid-water.

FRESHWATER RELATIVES

The only exclusively freshwater member of the Gadiformes order, the burbot is commonly found in cold rivers and lakes in the northern hemisphere. It prefers cold water, staying beneath the thermocline.

